

1. Why Uncertainty Matters in AI?

- Real-world environments are uncertain, noisy, partially observable and dynamic.
- Sensors provide incomplete or incorrect information.
- Actions can have unpredictable outcomes.
- We must make rational decisions under uncertainty, not just find a single "correct" answer.
- Probability theory provides a principled framework for representing and reasoning with uncertainty.

Key Idea: AI agents use probabilities to model uncertainty and to make good decisions even with incomplete or noisy information.

2. Limitations of Pure Logic

- Classical logic is deterministic: a statement is either true or false.
- It cannot naturally represent partial belief or uncertain information.
- In real problems, we often do not know the exact truth value of propositions.
- Logic does not handle noisy sensing, conflicting evidence or uncertain action outcomes.
- Probability generalizes logic by allowing degrees of belief between 0 and 1.

Example: In an uncertain world, the statement "It is raining" may be neither definitely true nor false; we might assign it a probability like 0.7.

3. Probability as a Degree of Belief

- Probability represents how strongly we believe a proposition or event will occur.
- Values range from 0 (impossible) to 1 (certain).
- Probabilities are based on evidence and updated as we get more information.
- A probability of 0.7 means we believe the event is likely, not that it will occur in 70% of cases for a single trial.
- Probability provides a consistent and quantitative way to combine uncertain information from multiple sources.

Key Idea: Probability is a numerical measure of rational belief under uncertainty, based on available evidence.

4. Basic Probability Axioms

Axioms (Kolmogorov)

- Non-negativity:** For any event A , $P(A) \geq 0$.
- Normalization:** $P(S) = 1$, where S is the sample space (some event must occur).
- Additivity:** If A and B are mutually exclusive ($A \cap B = \emptyset$), then $P(A \cup B) = P(A) + P(B)$.

From these axioms, we can derive other rules, such as: $P(\neg A) = 1 - P(A)$.

5. Events and Sample Space

- An experiment has a set of possible outcomes called the sample space, denoted by S .
- An event is any subset of the sample space.
- Events can be simple (single outcome) or compound (multiple outcomes).
- Important event operations:
 - Union ($A \cup B$): A or B (or both).
 - Intersection ($A \cap B$): A and B (both).
 - Complement ($\neg A$): not A .

Example: Rolling a fair die

Sample space: $S = \{1, 2, 3, 4, 5, 6\}$

Event A: even number = $\{2, 4, 6\}$

Event B: number $> 4 = \{5, 6\}$

$A \cup B = \{2, 4, 5, 6\}$

$A \cap B = \{6\}$

$\neg A = \{1, 3, 5\}$



6. Random Variables

- A random variable maps each outcome in the sample space to a numerical value.
- We use random variables to represent quantities of interest, such as the number of heads, a reward value, or a sensor reading.
- Notation:** capital letters (e.g., X, Y) denote random variables; lowercase (e.g., x) denote specific values.
- A random variable induces a probability distribution over its possible values.

Example: Let X be the number on a fair die.

x	1	2	3	4	5	6
$P(X=x)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

Here X is a discrete random variable.

7. Discrete vs. Continuous Variables

Discrete Random Variable

- Takes values from a countable set (finite or infinite).
- Described by a probability mass function (PMF): $P(X=x)$ for each value x .
- Must satisfy:
 - $P(X=x) \geq 0$ for all x .
 - $\sum_x P(X=x) = 1$.
- Examples: number of heads, number of customers, weather state.

Continuous Random Variable

- Takes values from a continuous range (e.g., real numbers).
- Described by a probability density function (PDF): $p(x)$.
- Must satisfy:
 - $p(x) \geq 0$ for all x .
 - $\int_{-\infty}^{\infty} p(x) dx = 1$.
- Examples: temperature, time to failure, position.

8. Probability Distributions

- A probability distribution specifies the probability of each possible value of a random variable.
- Discrete: given by a PMF $P(X=x)$.
- Continuous: given by a PDF $p(x)$ (probabilities are areas under the curve).
- Distributions summarize our knowledge about uncertain quantities and allow prediction, inference and decision making.

Example: Distribution of a biased coin

Let X be the number of heads in one toss, with $P(\text{heads}) = 0.7$ and $P(\text{tails}) = 0.3$.

x	0 (tails)	1 (heads)
$P(X=x)$	0.3	0.7

9. Joint Probability Distribution

- A joint distribution specifies the probability of every combination of values for a set of random variables.
- For two variables X and Y , the joint PMF is $P(X=x, Y=y)$.
- From the joint distribution we can compute marginal probabilities, e.g., $P(X=x) = \sum_y P(X=x, Y=y)$.
- Joint distributions capture relationships between variables (e.g., correlation or conditional independence).

Example: Joint distribution of two binary variables

$X \mid Y$	0	1	$P(X=x)$
0	0.2	0.3	0.5
1	0.1	0.4	0.5
$P(Y=y)$	0.3	0.7	1.0

1. Conditional Probability

- Measures the probability of an event given that another event has occurred.
- Defined as:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad (P(B) > 0)$$

- $P(A|B)$ is the probability of A given B.
- $P(A \cap B)$ is the probability of both A and B.

Example: If 30% of patients have a disease and 90% of diseased patients test positive, then $P(\text{Test}+ | \text{Disease}) = 0.9$.

2. Product Rule

- The joint probability of two events can be written using the product rule:

$$P(A \cap B) = P(A|B) P(B) \\ = P(B|A) P(A)$$

- Useful for decomposing joint probabilities into simpler conditional terms.
- Can be extended to sequences using the chain rule:

$$P(X_1, X_2, \dots, X_n) = \prod_{i=1}^n P(X_i | X_1, \dots, X_{i-1})$$

3. Bayes' Rule

- Relates $P(A|B)$ to $P(B|A)$:

$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$

- Allows us to update our belief in a hypothesis after observing evidence.
- Central to diagnostic reasoning.

Example: Probability of disease given a positive test result.

$$P(D|+) = \frac{P(+|D) P(D)}{P(+)}$$

4. Prior, Likelihood, Evidence, Posterior

Term	Notation	Meaning
Prior	$P(H)$	Initial belief in hypothesis before seeing evidence.
Likelihood	$P(E H)$	Probability of evidence if hypothesis is true.
Evidence	$P(E)$	Total probability of observing the evidence.
Posterior	$P(H E)$	Updated belief after seeing evidence.

5. Numerical Example: Medical Test

Example: A disease affects 1% of a population. A test for the disease has:

- $P(+|D) = 0.9$ (90% sensitive)
- $P(+|\neg D) = 0.05$ (5% false positive rate)

What is the probability a person has the disease given a positive test result?

Solution:

- Given:
 - $P(D) = 0.01$
 - $P(+|D) = 0.9$
 - $P(+|\neg D) = 0.05$
 - $P(\neg D) = 1 - 0.01 = 0.99$

- Use Bayes' rule:

$$P(D|+) = \frac{P(+|D) P(D)}{P(+|D) P(D) + P(+|\neg D) P(\neg D)} \\ = \frac{0.9 \times 0.01}{0.9 \times 0.01 + 0.05 \times 0.99} \\ = \frac{0.009}{0.009 + 0.0495} \\ = \frac{0.009}{0.0585} \approx 0.154$$

Even with a positive test, the probability of actually having the disease is only about 15.4% due to the low base rate.

6. Diagnostic Reasoning

- Use observations (evidence) to infer the most likely cause (hypothesis).
- Often use Bayes' rule to compute $P(H|E)$ for possible hypotheses.
- Common in medical diagnosis, fault detection, and classification.



Compute $P(D|+, S)$ using Bayes' rule and combine multiple pieces of evidence.

7. Base-Rate Effect

- People often ignore the base rate (prior probability) and focus only on the likelihood of the evidence.
- This can lead to overestimating rare events.
- Bayes' rule correctly accounts for both the base rate and the likelihood.

Example: In the medical test example, even though the test is 90% accurate, the low disease prevalence (1%) keeps the posterior probability low ($\approx 15.4\%$).

8. Independence

- Two events A and B are independent if knowing A does not change the probability of B (and vice versa).

$$P(A|B) = P(A) \quad (\text{independence})$$

$$\text{Equivalently: } P(A \cap B) = P(A) P(B)$$

Example: Whether it rains today is (roughly) independent of the outcome of a coin toss.

9. Conditional Independence

- Two events A and B are conditionally independent given a third event C if knowing A provides no additional information about B once C is known.

$$P(A|B, C) = P(A|C)$$

$$\text{Equivalently: } P(A \cap B|C) = P(A|C) P(B|C)$$

- Written as: $A \perp B | C$
- Common in probabilistic graphical models.

Example: In a medical diagnosis setting, symptoms S_1 and S_2 are often conditionally independent given the disease D.



Given D, observing S_1 gives no additional information about S_2 .

10. Summary Comparison

Concept	Definition / Formula	Key Idea
Conditional probability	$P(A B) = \frac{P(A \cap B)}{P(B)}$	Probability of A given B.
Product rule	$P(A \cap B) = P(A B) P(B)$	Decompose joint probabilities.
Bayes' rule	$P(A B) = \frac{P(B A) P(A)}{P(B)}$	Update beliefs using evidence.
Prior	$P(H)$	Initial belief.
Likelihood	$P(E H)$	Evidence given hypothesis.
Evidence	$P(E)$	Total probability of evidence.
Posterior	$P(H E)$	Updated belief.
Independence	$P(A B) = P(A)$	No influence between events.
Conditional independence	$P(A B, C) = P(A C)$	Independent given C.

"Bayes' rule lets us turn evidence into belief, but only when we remember the base rates."

— AIMA (Ch.13)

1. Full Joint Distributions

- A full joint distribution gives a probability for every possible complete assignment of values to all random variables.
- It fully specifies the uncertainty in a domain.
- From the joint distribution we can answer any probabilistic query by summing over relevant variables.

Key Idea: The full joint distribution is a complete specification of a probability model.

2. Example: Burglary-Earthquake-Alarm

- Variables: B = Burglary, E = Earthquake, A = Alarm (all Boolean).
- A full joint distribution has $2^3 = 8$ entries, listing $P(B, E, A)$ for each combination.

B	E	A	$P(B, E, A)$
T	T	T	0.001
T	T	F	0.000
T	F	T	0.094
T	F	F	0.005
F	T	T	0.010
F	T	F	0.890
F	F	T	0.001
F	F	F	0.999

These numbers are just for illustration. They sum to 1 over all 8 entries.

3. Marginalization

- To find the probability of a subset of variables, sum (marginalize) over the other variables.

$$P(A) = \sum_{b,e} P(b, e, A)$$

- For the burglary example:

$$\begin{aligned} P(A = T) &= \sum_{b,e} P(b, e, A = T) \\ &= P(T, T, T) + P(T, F, T) \\ &\quad + P(F, T, T) + P(F, F, T) \end{aligned}$$

Marginalization adds up the probabilities of all complete worlds that are consistent with the query.

4. Inference by Enumeration

- Answer queries using the full joint distribution by summing over hidden variables (enumeration).
- General query:

$$P(X|e) = \alpha \sum_y P(X, y, e)$$

where y are the hidden variables and α is a normalizing constant.

- Steps:
 - Fix values for query variables X and evidence e .
 - Sum over all possible assignments to hidden variables y .
 - Normalize so the probabilities sum to 1.

Example: Compute $P(B|A)$ by summing over the earthquake variable E .

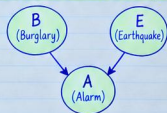
5. Predictive vs. Diagnostic Reasoning

- Predictive reasoning (causal):** infer effects from known causes (top-down).
 - Example: $B, E \rightarrow A$
 - Given $P(B), P(E)$, infer $P(A)$.
- Diagnostic reasoning (evidential):** infer causes from observed effects (bottom-up).
 - Example: $A \rightarrow B, E$
 - Given A , infer $P(B|A)$ and $P(E|A)$.

Predictive: cause $\rightarrow$ effect
Diagnostic: effect $\rightarrow$ cause

The same model supports both kinds of reasoning. The direction of inference depends on what variables we observe and what we want to query.

6. Causal Model for the Example



Joint factorization:

$$P(B, E, A) = P(B)P(E)P(A|B, E)$$

- B and E are independent (no direct arrow).
- A depends on both B and E .
- This compact representation is much smaller than listing all 8 joint probabilities explicitly for larger domains.

7. Intercausal Reasoning

- Two or more causes can independently influence the same effect.
- Observing the effect creates a dependence between the causes, even if they are independent a priori.
- This is called intercausal reasoning or explaining away.

Key Point: When A is observed, evidence for one cause (e.g. burglary) reduces the probability of the other cause (e.g. earthquake).

8. Explaining Away

- If the alarm has gone off, a burglary makes an earthquake less likely, because the alarm can be explained by the burglary.
- Without knowing anything:

B and E are independent:
 $P(B, E) = P(B)P(E)$.

- Given the alarm:

B and E become dependent:
 $P(B|E, A) < P(B|A)$
(seeing earthquake explains away the need for burglary).

Explaining away is a classic example of intercausal reasoning.

9. Why Full Joint Distributions are Expensive

- With n Boolean variables, the full joint distribution needs 2^n probabilities.
- The number of entries grows exponentially with the number of variables.
- Even for $n = 30$, we would need over one billion (2^{30}) numbers.
- Most entries will be very small and are hard to estimate from data.
- Using a Bayesian network with local conditional probabilities is much more compact, typically requiring a number of parameters that grows only linearly or polynomially with n (depending on the network structure).

Full joint distributions are conceptually simple but infeasible for large real-world domains.

10. Summary and Takeaways

- A full joint distribution specifies a probability for every complete world.
- Marginalization sums over variables to compute probabilities of interest.
- Inference by enumeration uses the joint distribution to answer queries.
- Predictive reasoning infers effects from causes; diagnostic reasoning infers causes from effects.
- Multiple independent causes of the same effect lead to intercausal reasoning.

- Explaining away: observing one cause reduces the probability of alternative causes.
- The burglary-earthquake-alarm example illustrates these ideas.
- Full joint distributions are expensive (exponential in n).
- Structured models such as Bayesian networks give a much more compact representation of the joint distribution.

"Probabilistic models let us reason with uncertain information about the world."

1. Continuous Variables

- Many real-world quantities are continuous and can take any value in a range.
- Examples: temperature, time, position, distance, height, sensor readings.
- We cannot list all possible values, so we use a probability density function (PDF) to describe uncertainty.
- The probability of an exact value is 0; we reason about intervals.

Key Idea: A continuous random variable is described by a probability distribution over a continuum of values.

Example: Temperature T can be any real value, e.g., 20.3°C , 20.31°C , ...

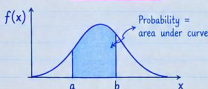
2. Probability Density Function (PDF)

- A probability density function $f(x)$ gives the relative likelihood of different values.
- $f(x) \geq 0$ for all x .
- The total area under the curve is 1.
- The probability of an interval $[a, b]$ is the area under the curve:

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

- The probability of a single value is 0:

$$P(X = x) = 0$$

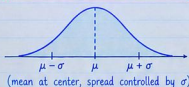


Example: If $f(x)$ is a normal distribution, $P(10 \leq X \leq 12)$ is the shaded area.

3. Normal Distribution

- A common continuous distribution for many real-world variables.
- Specified by a mean μ and standard deviation σ (variance σ^2).
- Bell-shaped and symmetric.
- Often a good model for sums of small effects (Central Limit Theorem).

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$



4. Mean, Variance and Expectation

- Mean (expected value) is the long-run average value of the variable.

$$E[X] = \int_{-\infty}^{\infty} x f(x) dx$$

- Variance measures the spread of the distribution.

$$\text{Var}(X) = E[(X - \mu)^2]$$

$$\sigma^2 = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx$$

- Standard deviation $\sigma = \sqrt{\text{Var}(X)}$.
- Expectation is a weighted average of all possible values.
- For a function $g(x)$:

$$E[g(x)] = \int_{-\infty}^{\infty} g(x) f(x) dx$$

Example: If $X \sim N(\mu, \sigma^2)$ then

$$E[X] = \mu \quad \text{Var}(X) = \sigma^2$$

5. Probability as a Degree of Belief

- Probability represents a rational degree of belief, not just frequency.
- It quantifies uncertainty based on available evidence.
- Probabilities are subjective but should follow the rules of probability (0 to 1, add to 1, etc.).
- We update probabilities when we obtain new evidence (e.g., using Bayes' rule).

Key Idea: Probability is a consistent measure of how strongly we believe a proposition is true, given the evidence.

6. Common Probability Mistakes

- Confusing $P(A|B)$ and $P(B|A)$:** $P(\text{doctor}|\text{disease})$ is not the same as $P(\text{disease}|\text{doctor})$.
- Base-rate neglect:** ignoring the prior probability of an event.
- Treating small probabilities as impossible:** rare events can still happen.
- Assuming independence:** when variables are actually related.
- Overconfidence:** assigning probabilities too close to 0 or 1.
- Confusing correlation with causation.
- Ignoring uncertainty in sensor readings and models.

Note: Probabilistic reasoning helps avoid these mistakes by explicitly representing uncertainty.

7. Practical Application: Diagnosis

- Use probabilities to reason about a patient's disease given test results.
- Combine prior probability (prevalence) with test accuracy using Bayes' rule.

Example: A disease affects 1% of the population. A test is 99% accurate (true positive rate = 0.99, false positive rate = 0.01). What is $P(\text{disease} | \text{positive})$?

$$P(D|+) = \frac{P(+|D) P(D)}{P(+|D) P(D) + P(+|\neg D) P(\neg D)}$$

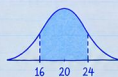
$$P(D|+) \approx 0.50 \quad (\text{about } 50\%)$$

Even with a very accurate test, the probability can be much lower than expected due to the low base rate.

8. Practical Application: Forecasting

- Predict future values such as weather, demand, or stock prices.
- Model the quantity as a probability distribution (often normal or other parametric forms).
- Use the distribution to answer queries like "What is the probability it will rain tomorrow?" or "What is the expected demand next week?"

Example: Tomorrow's temperature T is modeled as $T \sim N(20, 4^\circ)$. What is $P(16 \leq T \leq 24)$?



$$P(16 \leq T \leq 24) \approx 0.68 \quad (\text{about } 68\%)$$

9. Practical Application: Localization

- A robot's position is uncertain due to noisy sensors and motion.
- Represent the robot's location with a continuous distribution (e.g., Gaussian over position).
- Update the distribution using motion and sensor measurements.
- Used in tracking, robot navigation, and autonomous vehicles.

Example: A robot's 1D position x is modeled as $x \sim N(10, 2^2)$. After a noisy measurement it updates to $x \sim N(11, 1^2)$, a more certain estimate (smaller variance).

1. Key Definitions (Review)

- Random variable:** variable with uncertain value, described by a probability distribution.
- Outcome:** a specific assignment of values to a set of variables.
- Event:** a set of outcomes.
- Probability:** $P(X = x) \in [0, 1]$, with $\sum_x P(X = x) = 1$.
- Joint probability:** $P(X, Y)$ over multiple variables.
- Conditional probability:** $P(X|Y) = \frac{P(X, Y)}{P(Y)}$ (if $P(Y) > 0$).
- Independence:** $X \perp Y$ if $P(X|Y) = P(X)$.
- Bayes' rule:** $P(X|E) = \frac{P(E|X)P(X)}{P(E)}$.

Key Idea: Uncertainty is represented and manipulated using probability theory.

2. Formula Recap

Normalization : $\sum_x P(X = x) = 1$
 Complement : $P(\neg A) = 1 - P(A)$
 Addition rule : $P(A \vee B) = P(A) + P(B) - P(A \wedge B)$
 Conditional prob: $P(A|B) = \frac{P(A, B)}{P(B)}$
 Product rule : $P(A, B) = P(A|B)P(B)$
 Bayes' rule : $P(A|B) = \frac{P(B|A)P(A)}{P(B)}$
 Total probability: $P(A) = \sum_i P(A|B_i)P(B_i)$
 Independence : $P(A|B) = P(A)$ ($A \perp B$)
 Marginalization: $P(X) = \sum_y P(X, Y = y)$
 Expectation : $E[X] = \sum_x x P(X = x)$

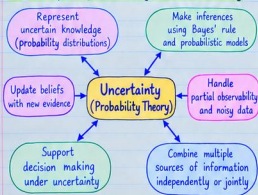
These formulas let us represent, infer, and reason under uncertainty.

3. Prior, Likelihood, Posterior, Joint, Marginal, Conditional

Quantity	Notation	Meaning	Example
Prior probability	$P(H)$	Initial belief before evidence	$P(\text{Flu}) = 0.01$
Likelihood	$P(E H)$	How likely evidence is if hypothesis true	$P(\text{Cough} \text{Flu}) = 0.9$
Posterior probability	$P(H E)$	Updated belief after evidence	$P(\text{Flu} \text{Cough}) = 0.41$
Joint probability	$P(X, Y)$	Probability of both variables	$P(\text{Cloudy}, \text{Rain}) = 0.2$
Marginal probability	$P(X)$	Probability of one variable (sum out others)	$P(\text{Rain}) = \sum_c P(\text{Rain}, c)$
Conditional probability	$P(X Y)$	Probability of X given Y	$P(\text{Rain} \text{Cloudy}) = 0.6$

Posterior $\propto$ Likelihood $\times$ Prior (Bayes' rule)

4. Concept Map: Reasoning Under Uncertainty



Uncertainty connects representation, inference, learning and decision making.

5. Common Exam Questions

- Compute a conditional probability using Bayes' rule.
- Decide whether variables are independent given a joint distribution.
- Marginalize to find the distribution of a single variable.
- Use the total probability rule in a situation with multiple causes.
- Construct a joint distribution from a graphical model structure.
- Update a belief state after new evidence.
- Compare two hypotheses using likelihoods.
- Explain the difference between prior, likelihood and posterior.

Example: A medical test is 90% accurate if a disease is present and 5% false positive. The disease prevalence is 1%. What is the probability of disease given a positive test?

6. Worked Example (Quick Review)

Example: Suppose $P(\text{Disease}) = 0.01$,
 $P(\text{Test}+|\text{Disease}) = 0.90$,
 $P(\text{Test}+|\neg\text{Disease}) = 0.05$.
 Find $P(\text{Disease}|\text{Test}+)$.

Solution:

$$P(D|+) = \frac{P(+|D)P(D)}{P(+)}$$

$$P(+) = P(+|D)P(D) + P(+|\neg D)P(\neg D)$$

$$= 0.90(0.01) + 0.05(0.99)$$

$$= 0.009 + 0.0495 = 0.0585$$

$$P(D|+) = \frac{0.90(0.01)}{0.0585} = \frac{0.009}{0.0585} \approx 0.154$$

So there is about a 15.4% chance of disease given a positive test.

7. Graphical Models Perspective (Preview)

- A joint distribution can be compactly represented using a Bayesian network (directed graphical model).
- Nodes: random variables.
- Edges: conditional dependencies.
- Each node has a conditional probability table (CPT).
- Joint distribution factorizes as:

$$P(X_1, \dots, X_n) = \prod_i P(X_i | \text{Parents}(X_i))$$
- Independence relations can be read from the graph (conditional independence).
- Graphical models make reasoning and learning scalable for large domains.

Bayesian networks are a key tool for probabilistic reasoning (Chapter 14).

8. Chapter Takeaways

- Probability theory provides a rigorous framework for representing uncertainty.
- We can combine prior knowledge with evidence using Bayes' rule.
- Joint, marginal and conditional probabilities allow flexible queries.
- Independence assumptions simplify representation and inference.
- Uncertainty is essential for handling noisy, incomplete and ambiguous real-world information.
- Probabilistic representations support reasoning, learning and decision making.
- Bayesian networks, introduced later, provide a structured way to represent and reason with complex domains.

9. Bridge to Chapter 14: Probabilistic Reasoning

- Chapter 13 introduced the foundations of probability and basic inference.
- Chapter 14 develops probabilistic reasoning in more detail, focusing on Bayesian networks.
- We will learn:
 - how to construct and use Bayesian networks,
 - inference algorithms (exact and approximate),
 - learning network structures and parameters,
 - applications in diagnosis, prediction and decision making.

Uncertainty is the foundation. Chapter 14 shows how to reason efficiently with uncertainty using graphical models.

10. Summary: Key Points to Remember

- Uncertainty is fundamental in real-world AI.
- Use probability theory to represent and reason.
- Bayes' rule updates beliefs with evidence.
- Distinguish prior, likelihood, posterior, joint, marginal and conditional probabilities.
- Graphical models provide a compact and scalable representation.

"Uncertainty is not a bug in the real world; it is a fundamental part of it."